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Improvements from Task 1:

VLAN Implementation for the Network:

- Implemented VLAN 10 (Main) and VLAN 20 (Guest).

Internet Simulation:

- Built a custom internet simulation.

Firewall/ACL Implementation:

- Implemented ACL 110 to deny guest VLAN access to the main network.
- Allow internet access for both networks.

DHCP and IP Planning:

- DHCP pools were implemented per VLAN.
- Static IP are assigned to key devices (e.g TV, Router, Gaming Console).

Coverage and Hardware Placement:

- Access Points were positioned to provide floor-based coverage (2.4 GHz up, 5 GHz down).

Documentation & Screenshots

IP Interface on Router for VLANs:

```
HomeRouter>show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 unassigned      YES manual up          up
GigabitEthernet0/0.10 192.168.1.1    YES manual up          up
GigabitEthernet0/0.20 192.168.2.1    YES manual up          up
GigabitEthernet0/1    203.0.113.1    YES manual up          up
Vlan1               unassigned      YES unset  administratively down down
HomeRouter>|
```

Note: 203.0.113.1 is used for WAN simulation.

IP DHCP Pool for both the Main and Guest Networks:

```
HomeRouter>show ip dhcp pool
```

```
Pool MAIN_WIFI_POOL :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)          : 0 / 0
  Total addresses                    : 254
  Leased addresses                   : 8
  Excluded addresses                 : 2
  Pending event                     : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  192.168.1.1        192.168.1.1 - 192.168.1.254    8 / 2 / 254

Pool GUEST_WIFI_POOL :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)          : 0 / 0
  Total addresses                    : 254
  Leased addresses                   : 5
  Excluded addresses                 : 2
  Pending event                     : none

  1 subnet is currently in the pool
  Current index      IP address range      Leased/Excluded/Total
  192.168.2.1        192.168.2.1 - 192.168.2.254    5 / 2 / 254
HomeRouter>
```

DHCP leases from both pools:

```
HomeRouter>show ip dhcp binding
```

IP address	Client-ID/ Hardware address	Lease expiration	Type
192.168.1.102	0001.9790.2CBB	--	Automatic
192.168.1.101	00D0.D32B.B36E	--	Automatic
192.168.1.100	000B.BE42.3023	--	Automatic
192.168.1.104	00D0.9721.B4C7	--	Automatic
192.168.1.103	00E0.F7E7.6300	--	Automatic
192.168.1.105	0030.A39A.C61E	--	Automatic
192.168.1.106	0001.4230.427B	--	Automatic
192.168.1.107	0007.EC26.1528	--	Automatic
192.168.2.13	00E0.B098.0B2B	--	Automatic
192.168.2.10	00E0.F7BB.933D	--	Automatic
192.168.2.11	00E0.8F2D.55C3	--	Automatic
192.168.2.12	00E0.F753.70E4	--	Automatic
192.168.2.14	000C.CF3D.4E6B	--	Automatic

```
HomeRouter>
```

Access Point Configurations:

Port 1

Port Status

☒ On

SSID

Network5GHz

5 GHz Channel

36

Coverage Range (meters)

140.00

Authentication

☐ Disabled

☐ WEP

☐ WPA-PSK

☒ WPA2-PSK

WEP Key

PSK Pass Phrase

User ID

Password

7o7_H0m3N3tw0rk@2025

Encryption Type

AES

Note: The SSID would be hidden in a real-world application; however, Cisco Packet Tracer does not allow the application to save without it.

Port 1

Port Status

☒ On

SSID

GuestNetwork

2.4 GHz Channel

6

5 GHz Channel

112

Coverage Range (meters)

250.00

Authentication

☐ Disabled

☐ WEP

☐ WPA-PSK

☒ WPA2-PSK

WEP Key

PSK Pass Phrase

User ID

Password

:5t_W1F1@3dg3H!!!#2025

Encryption Type

AES

Switch VLAN setup:

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24
10	MAIN_WIFI	active	Gig0/2
20	GUEST_WIFI	active	Fa0/1
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>
```

Trunk port configuration allowing VLAN 10 & 20 for Main and Guest:

```
Switch>show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Gig0/1    on        802.1q         trunking      1

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20

Switch>
```

Access List configuration - Guest network traffic is blocked from Main:

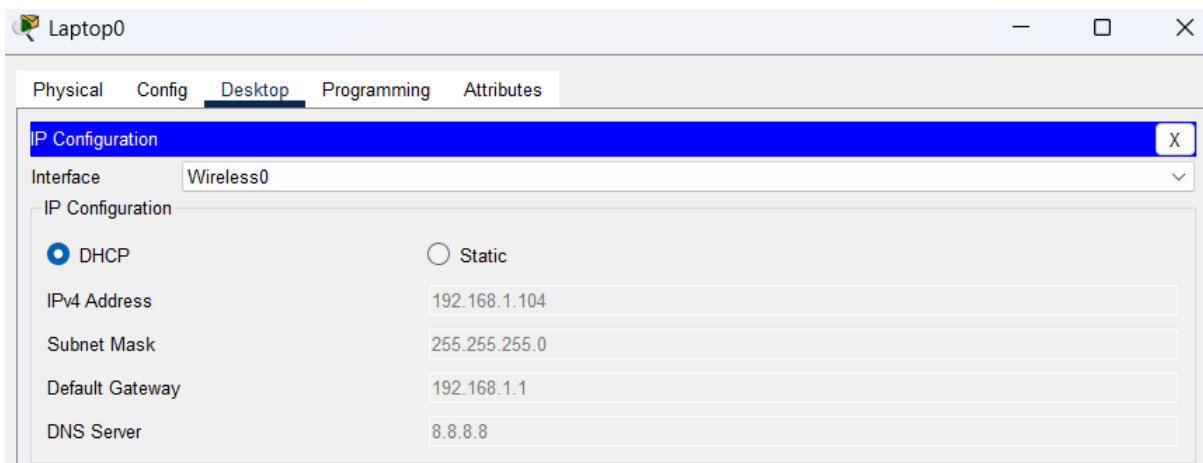
```
HomeRouter>enable
HomeRouter#show access-lists
Extended IP access list 110
  10 deny ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
  20 permit ip any any (10 match(es))

HomeRouter#
```

ACL 110 is applied inbound to the VLAN 20 Guest Network, preventing access to the Main Network while still allowing internet access:

```
HomeRouter#show run | include access-group
ip access-group 110 in
```

Device IP Configuration Main, Guest, & Internet PC:



Home Wireless Networks: TASK 2

GUEST_LAPTOP_1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface Wireless0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.2.12

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 8.8.8.8

Internet_PC

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 8.8.8.8

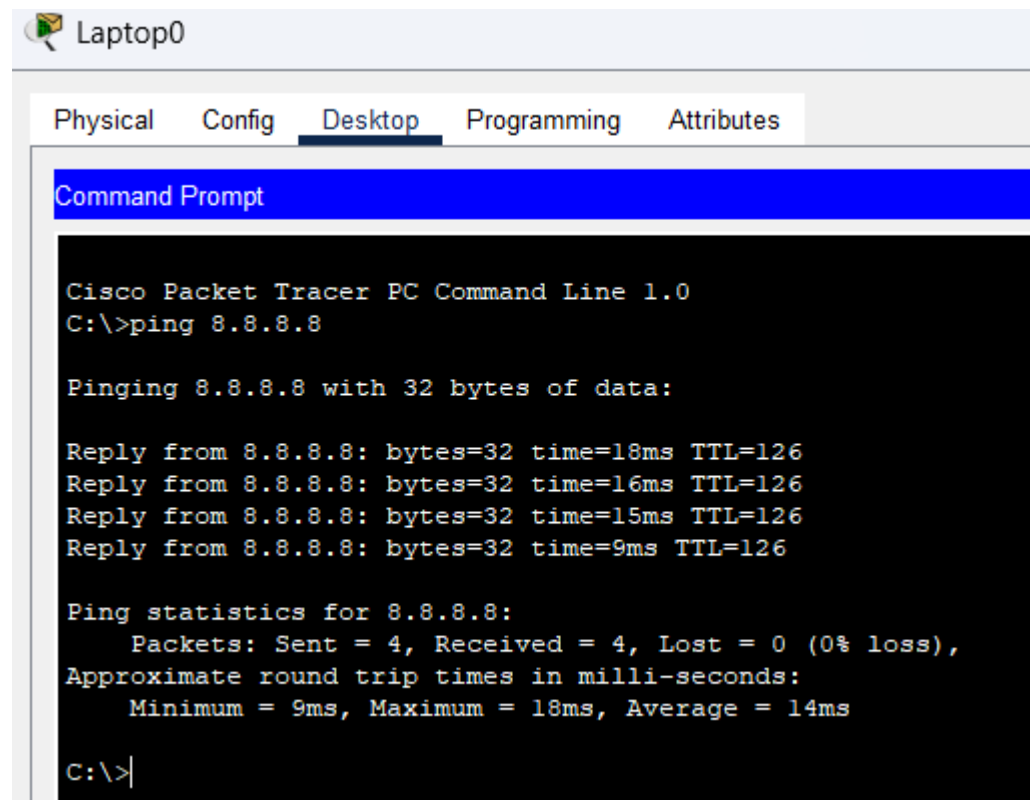
Subnet Mask 255.255.255.0

Default Gateway 8.8.8.1

DNS Server 0.0.0.0

Network Ping Testing:

Main to Internet:



The screenshot shows the Cisco Packet Tracer PC Command Line interface for a device named 'Laptop0'. The 'Desktop' tab is selected. The command prompt displays the output of a 'ping 8.8.8.8' command. The output indicates that four packets were sent and received successfully with 0% loss. The round trip times are: 18ms, 16ms, 15ms, and 9ms, with an average of 14ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

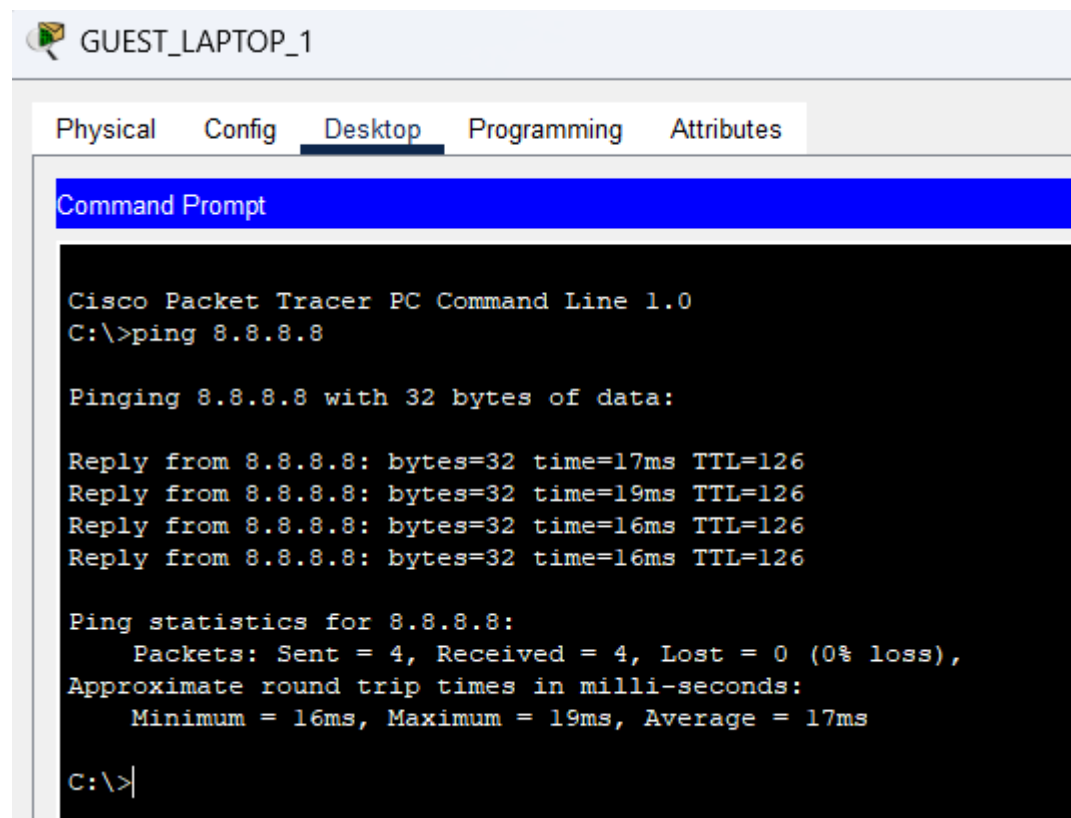
Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=18ms TTL=126
Reply from 8.8.8.8: bytes=32 time=16ms TTL=126
Reply from 8.8.8.8: bytes=32 time=15ms TTL=126
Reply from 8.8.8.8: bytes=32 time=9ms TTL=126

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 18ms, Average = 14ms

C:\>
```

Guest to Internet:



The screenshot shows the Cisco Packet Tracer PC Command Line interface for a device named 'GUEST_LAPTOP_1'. The 'Desktop' tab is selected. The command prompt displays the output of a 'ping 8.8.8.8' command. The output indicates that four packets were sent and received successfully with 0% loss. The round trip times are: 17ms, 19ms, 16ms, and 16ms, with an average of 17ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

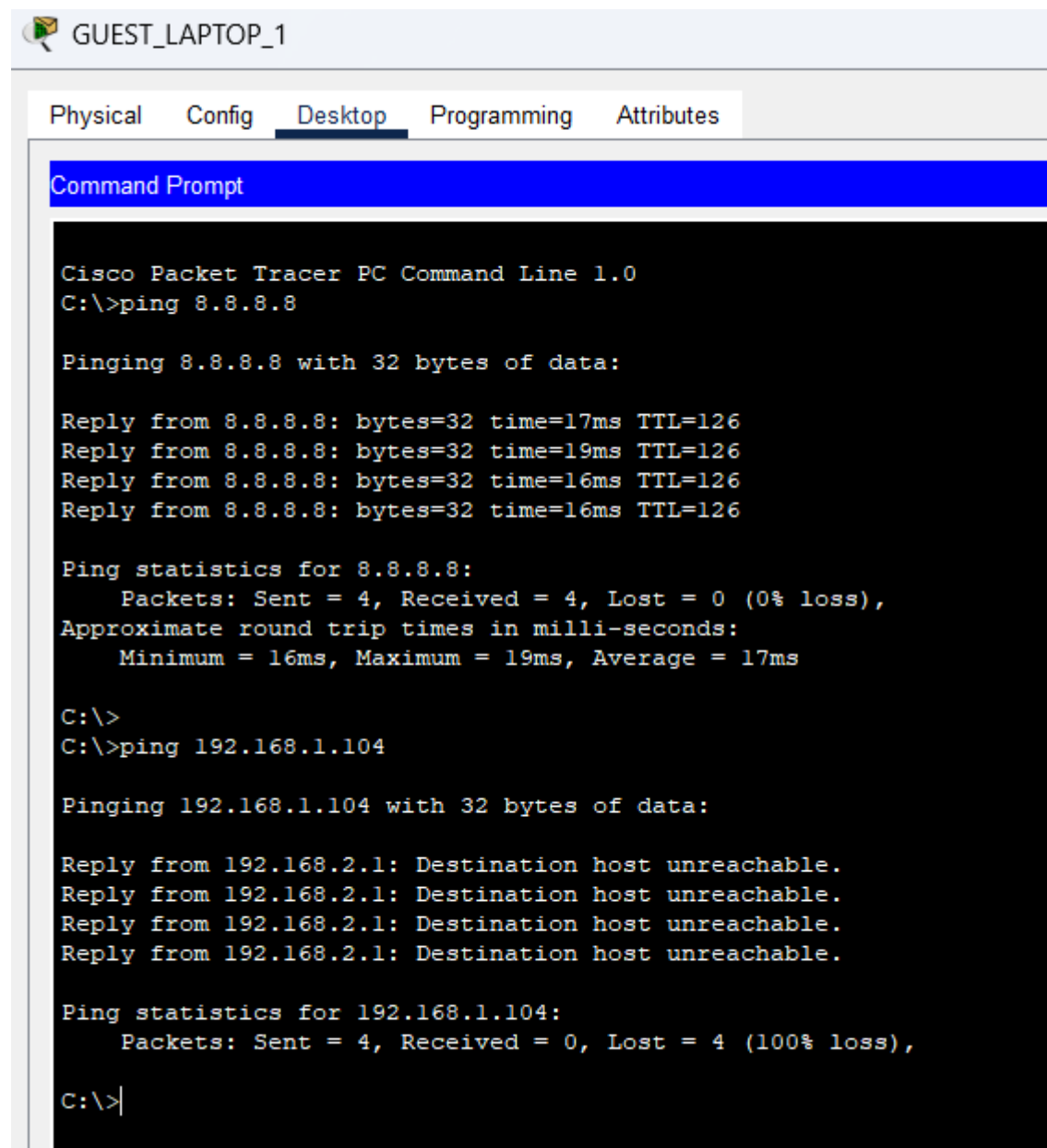
Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=17ms TTL=126
Reply from 8.8.8.8: bytes=32 time=19ms TTL=126
Reply from 8.8.8.8: bytes=32 time=16ms TTL=126
Reply from 8.8.8.8: bytes=32 time=16ms TTL=126

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 19ms, Average = 17ms

C:\>
```

Guest to Main:



```
GUEST_LAPTOP_1

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=17ms TTL=126
Reply from 8.8.8.8: bytes=32 time=19ms TTL=126
Reply from 8.8.8.8: bytes=32 time=16ms TTL=126
Reply from 8.8.8.8: bytes=32 time=16ms TTL=126

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 19ms, Average = 17ms

C:\>
C:\>ping 192.168.1.104

Pinging 192.168.1.104 with 32 bytes of data:

Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.
Reply from 192.168.2.1: Destination host unreachable.

Ping statistics for 192.168.1.104:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```


MAC Address Filtering:

To further enhance wireless security on the main network (5GHz), MAC address filtering was initially intended to be implemented. This feature only allows specific trusted devices to connect to the wireless network based on their hardware (MAC) address.

However, due to Cisco Packet Tracer limitations, MAC address filtering is not available within the standard AccessPoint-PT-AC. On the other hand, in a real-world scenario, MAC filtering could be implemented with the following:

- Filter Mode: Allow only listed MAC addresses.
- Device Examples: Laptop0 (MAC: 00D0.9721.B4C7), Laptop1 (MAC: 00D0.D32B.B36E).